Study Memo (Condensed): Can Machine Learning Beat Classical Volatility Models?

François Schmitt (23035010523), Trimester 9 Term Project

Concepts in one line each

Volatility: the conditional standard deviation σ_t of returns; drives option prices, VaR, margins, position sizes.

Task: at the close of day $t-1$, forecast day t's variance $\hat{\sigma}_t^2$ using information up to $t-1$ only.

Latent volatility: never observed on any day; forecasts are scored against the proxy r_t^2, unbiased but very noisy.

Volatility clustering: ACF of returns is flat, ACF of $|r_t|$ decays slowly and stays significant; magnitude is forecastable, direction is not.

Leverage effect: negative returns raise future volatility more than positive ones; encoded by GJR (threshold term) and EGARCH (signed log-variance term).

Walk-forward: the final 30% of days is the shared test window; every forecast uses past data only; 4,650 returns 2008–2026, test = 1,395 days (2020-12-07 to 2026-06-29, holding the 2022 bear and the April-2025 tariff shock; COVID is in train).

Refit-and-roll: GARCH-family parameters re-fit by MLE every 22 days on an expanding window; the recursion rolls forward daily on observed returns in between; recursions verified against the `arch` package source.

Naive / Rolling / EWMA: r_{t-1}^2; variance of last 21 days; RiskMetrics $\hat{\sigma}_t^2 = 0.94\,\hat{\sigma}_{t-1}^2 + 0.06\,r_{t-1}^2$.

GARCH(1,1): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$; three parameters, persistence via $\alpha + \beta \approx 1$.

Features (shared): return and squared-return lags 1–10, realized variance over 5 and 21 days (HAR horizons), 5-day mean $|r|$.

Target (shared): $y_t = \ln(\frac{1}{5}\sum_{j=0}^{4} r_{t+j}^2 + \varepsilon)$: forward 5-day smoothed log variance; needs a 5-day training buffer since y_t is only known at $t+4$.

XGBoost: 400 shallow trees, refit from scratch every 22 days (same cadence as GARCH); warm-starting rejected for reproducibility.

LSTM: 1 layer, 32 units, reads 20 days of standardized features; train-only scaling, validation from end of train, early stopping (stopped epoch 37); sequences labelled at their last row.

RMSE / MAE: symmetric point-accuracy on the volatility scale; interpretable, but blind to the asymmetry a risk manager cares about.

QLIKE: mean$(p/v - \ln(p/v) - 1)$ with $p = r_t^2$, $v = \hat{\sigma}_t^2$; ranking robust to proxy noise (Patton 2011) and punishes under-prediction of variance; the headline metric.

Naive QLIKE blow-up: p/v diverges when a near-zero forecast meets movement; hence naive scores 5.4×10^3 vs ≈ 1.5 for the rest.

Diebold–Mariano: tests whether the mean daily loss gap between two models is real; Harvey–Leybourne–Newbold small-sample correction; negative statistic favours the first model.

Reproducibility: fixed seeds, one shared split, five independent heavy fits in parallel processes; full study re-runs in about one minute.

Results in five lines

EGARCH(1,1) wins QLIKE (1.5042), statistically tied with GJR-GARCH (1.5091, $p = 0.47$).

Leverage now matters: both beat plain GARCH(1,1) decisively ($p < 0.0001$); on the 2008–2024 window plain GARCH had won and the edge was insignificant.

LSTM is third (1.5287), ahead of plain GARCH (1.5569), and statistically tied with the winner ($p = 0.19$).

Learned models take point accuracy (LSTM best RMSE 0.00686, XGBoost best MAE 0.00496) but trail on QLIKE: smoothed targets react late at spikes.

Verdict: no learned model outranks the leverage GARCH family where it matters; the ML case is narrower than before, not closed.

Quick-fire Q&A

Q. Why forecast volatility, not returns? A. Returns have a flat ACF (unpredictable direction); $|r_t|$ has a slowly decaying ACF (predictable magnitude). Only the second is a forecasting problem.

Q. How do you score forecasts of something never observed? A. Against the unbiased proxy r_t^2 with QLIKE, whose model ranking is provably robust to proxy noise.

Q. Why QLIKE over RMSE? A. Robust ranking under a noisy proxy, and asymmetric: under-predicting variance in a crash is punished hardest, matching real risk costs. Here the learned models win RMSE/MAE yet lose QLIKE.

Q. Why not train ML on r_t^2 directly? A. Too noisy; squared-error learners collapse toward a constant. Hence the log 5-day-forward smoothed target, paid for with a 5-day training buffer.

Q. Why walk-forward, not cross-validation? A. Random splits train on the future. Walk-forward reproduces deployment: past-only information, periodic refits, one fixed out-of-sample tail.

Q. Monthly refits but daily forecasts: how? A. Parameters are re-estimated every 22 days; the fitted recursion updates the variance state daily from observed returns, exactly as the model equation prescribes.

Q. Why did the winner change when the sample was extended? A. The new test window holds two asymmetric crashes (2022, April 2025); the leverage term goes from insignificant to decisive ($p < 0.0001$). Model rankings are window-dependent; that is a finding, not a flaw.

Q. Is the LSTM good, then? A. Competitive: third place, best RMSE, statistically tied with the winner ($p = 0.19$). But tied is not better; the supported claim stays “no learned model outranks the leverage GARCH family".

Q. Why does the LSTM win RMSE but lose QLIKE? A. Smoothed target, symmetric training loss: accurate on average, one step late at eruptions, and QLIKE charges precisely for late under-prediction.

Q. Why is naive's QLIKE 5,440? A. The loss ratio p/v explodes whenever yesterday's flat return makes $v \approx 0$ and today moves. Property of the loss, visible only because the metric is right.

Q. What does the DM test add over the leaderboard? A. Error bars: it converts a QLIKE gap into a significance statement using the autocorrelation-consistent variance of daily loss differentials (with the HLN correction).

Q. Anti-leakage guards in the LSTM? A. Last-row sequence labels, train-only standardization, validation carved from the end of train, 5-day target buffer.

Q. Biggest lesson? A. The harness beats the model: leak-free evaluation and the choice of metric changed conclusions more than any architecture or hyperparameter; "better" is undefined until loss and window are fixed.

Code and results: `https://github.com/iitgF/T9`. *Full walkthrough:* `memo_standard.tex`.